

Instructions

This mock mirrors the Tredence data-science assessment: aptitude, verbal, statistics, ML and CS MCQs plus SQL and Python.

Attempt in one ~2-hour sitting. This is Mock 3 of 3; each uses a different question set.

Section 1 - Aptitude & Reasoning

Q1. If 60% passed and 80 failed, total students:

(A) 160

(B) 200 ✓

(C) 180

(D) 240

Answer: (B) 200

Authenticity: Community Reported

Q2. SI on Rs. 2000 at 5% for 4 years:

(A) Rs. 300

(B) Rs. 400 ✓

(C) Rs. 500

(D) Rs. 450

Answer: (B) Rs. 400

Authenticity: Community Reported

Q3. Sales (Rs. lakh): 2019=40,2020=50,2021=60,2022=80. Total over 4 years:

(A) 210

(B) 220

(C) 230 ✓

(D) 240

Answer: (C) 230

Authenticity: Research-Backed Company Style

Q4. Next term: 1, 3, 6, 10, 15, ?

(A) 18

(B) 20

(C) 21 ✓

(D) 24

Answer: (C) 21

Authenticity: Frequently Repeated

Q5. If $A=1, B=2, \dots$, value of SUN:

(A) 48

(B) 54 ✓

(C) 50

(D) 45

Answer: (B) 54

Authenticity: Community Reported

Section 2 - Verbal

Q1. Antonym of SCARCE:

(A) Rare

(B) Abundant ✓

(C) Limited

(D) Few

Answer: (B) Abundant

Authenticity: Community Reported

Q2. Choose the correct sentence.

(A) He is one of the best player.

(B) He is one of the best players. ✓

(C) He is one of best player.

(D) He is a best players.

Answer: (B) He is one of the best players.

Authenticity: Community Reported

Section 3 - Statistics & Probability

Q1. Find the mode of 2, 3, 3, 5, 7, 3, 8.

(A) 2

(B) 3 ✓

(C) 5

(D) 7

Answer: (B) 3

Authenticity: Community Reported

Q2. Which measure is most affected by outliers?

(A) Mode

(B) Median

(C) Mean ✓

(D) Range

Answer: (C) Mean

Authenticity: Frequently Repeated

Q3. The interquartile range (IQR) is:

(A) $Q3 - Q1$ ✓

(B) $Q3 + Q1$

(C) $\text{Max} - \text{Min}$

(D) $Q2 - Q1$

Answer: (A) $Q3 - Q1$

Authenticity: Community Reported

Q4. Expected value of a single fair die roll:

(A) 3

(B) 3.5 ✓

(C) 4

(D) 2.5

Answer: (B) 3.5

Authenticity: Frequently Repeated

Q5. Sum of probabilities of all outcomes in a sample space:

(A) 0

(B) 0.5

(C) 1 ✓

(D) Depends

Answer: (C) 1

Authenticity: Community Reported

Q6. Two coins tossed; P(at least one head):

(A) $1/4$

(B) $1/2$

(C) $3/4$ ✓

(D) 1

Answer: (C) $3/4$

Authenticity: Research-Backed Company Style

Q7. A Type II error is:

(A) Rejecting a true null

(B) Failing to reject a false null hypothesis ✓

(C) A measurement error

(D) A rounding error

Answer: (B) Failing to reject a false null hypothesis

Authenticity: Frequently Repeated

Q8. For a normal distribution, ~what percent lies within two SD of the mean?

(A) 68%

(B) 90%

(C) 95% ✓

(D) 99.7%

Answer: (C) 95%

Authenticity: Community Reported

Q9. As sample size increases, the standard error of the mean:

(A) Increases

(B) Decreases ✓

(C) Stays constant

(D) Becomes negative

Answer: (B) Decreases

Authenticity: Community Reported

Section 4 - Machine Learning

Q1. Supervised learning requires:

(A) Unlabeled data

(B) Labeled data ✓

(C) No data

(D) Only test data

Answer: (B) Labeled data

Authenticity: Frequently Repeated

Q2. Random Forest is an ensemble of:

(A) K-means runs

(B) Decision trees ✓

(C) Neural nets

(D) SVMs

Answer: (B) Decision trees

Authenticity: Community Reported

Q3. Boosting is an ensemble method that:

(A) Trains models independently in parallel

(B) Builds models sequentially, each correcting prior errors ✓

(C) Reduces features

(D) Clusters data

Answer: (B) Builds models sequentially, each correcting prior errors

Authenticity: Community Reported

Q4. Recall (sensitivity) is:

(A) $TP/(TP+FP)$

(B) $TP/(TP+FN)$ ✓

(C) $TN/(TN+FN)$

(D) $FP/(FP+TN)$

Answer: (B) $TP/(TP+FN)$

Authenticity: Frequently Repeated

Q5. Purpose of a train-test split:

(A) Speed up training

(B) Evaluate on unseen data ✓

(C) Balance classes

(D) Remove outliers

Answer: (B) Evaluate on unseen data

Authenticity: Frequently Repeated

Q6. Adjusted R-squared differs from R-squared because it:

(A) Always increases with more features

(B) Penalizes adding non-useful predictors ✓

(C) Ignores residuals

(D) Equals R-squared always

Answer: (B) Penalizes adding non-useful predictors

Authenticity: Community Reported

Q7. Output activation for multi-class classification:

(A) ReLU

(B) Softmax ✓

(C) Tanh

(D) Linear

Answer: (B) Softmax

Authenticity: Community Reported

Q8. One-hot encoding handles:

(A) Missing numerics

(B) Categorical variables ✓

(C) Outliers

(D) Imbalance

Answer: (B) Categorical variables

Authenticity: Frequently Repeated

Q9. Dropout in a neural network is used to:

(A) Speed up I/O

(B) Reduce overfitting ✓

(C) Increase parameters

(D) Normalize inputs

Answer: (B) Reduce overfitting

Authenticity: Community Reported

Section 5 - CS Fundamentals

Q1. Average time complexity of hash table lookup:

(A) $O(1)$ ✓

(B) $O(n)$

(C) $O(\log n)$

(D) $O(n \log n)$

Answer: (A) $O(1)$

Authenticity: Community Reported

Q2. The four pillars of OOP are encapsulation, abstraction, inheritance and:

(A) Compilation

(B) Polymorphism ✓

(C) Recursion

(D) Iteration

Answer: (B) Polymorphism

Authenticity: Community Reported

Section 6 - SQL

Problem: Employees Above Department Average

List employees earning above their own department's average, ordered by name.

```
SELECT name, salary
FROM staff s
WHERE salary > (SELECT AVG(salary) FROM staff WHERE dept = s.dept)
ORDER BY name;
```

Expected Output

```
name|salary
B|60000
C|90000
```

Section 7 - Python

Problem 1: FizzBuzz

Difficulty: **Easy** | Pattern: **Simulation / Modulo** | Authenticity: **Frequently Repeated**

Problem Statement

Print $1..n$; multiples of 3 -> Fizz, of 5 -> Buzz, of both -> FizzBuzz. Space-separated.

Input Format

A single integer n .

Output Format

The FizzBuzz sequence on one line.

Constraints

$1 \leq n \leq 10^4$

Sample Input

15

Sample Output

1 2 Fizz 4 Buzz Fizz 7 8 Fizz Buzz 11 Fizz 13 14 FizzBuzz

Python Solution

```
def fizzbuzz(n):
    out = []
    for i in range(1, n + 1):
        if i % 15 == 0:
            out.append('FizzBuzz')
        elif i % 3 == 0:
            out.append('Fizz')
        elif i % 5 == 0:
            out.append('Buzz')
        else:
            out.append(str(i))
    return ' '.join(out)

print(fizzbuzz(15))
```

Step-by-Step Explanation

Check 15 first, then 3, then 5, else the number. A very common warm-up coding question.

Dry Run

3->Fizz, 5->Buzz, 15->FizzBuzz

Time Complexity: **O(N)** | Space Complexity: **O(N)**

Tredence

Full-Length Mock Test 3

Applied Role(s):

Analyst - Data Scientist

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

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Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.